Lecture Notes on

Exoplanet Detection with Real Data

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These notes summarize the 6th Astrophysics Club meeting, where we used real TESS data and basic machine learning techniques to detect exoplanets using the Transit Method.

Contents

1 Meeting Objectives

- Review what an exoplanet is.
- Understand the Transit Method using real star data.
- Use Python to automatically detect transit dips.
- Introduce basic machine learning logic for pattern detection.

2 What is an Exoplanet?

An exoplanet (extrasolar planet) is any planet orbiting a star outside our Solar System.

Most exoplanets are far too faint to be directly observed. Instead, we detect them indirectly.

3 The Transit Method

When a planet passes in front of its star (from our perspective):

- It blocks a small fraction of starlight.
- We observe a temporary dip in brightness.
- This dip is called a **transit**.

Key measurable quantities:

- Transit depth $\rightarrow$ planet size.
- Time between transits $\rightarrow$ orbital period.
- Transit duration $\rightarrow$ orbital geometry.

Transit depth approximation:

$$\frac{\Delta F}{F} \approx \left(\frac{R_p}{R_*}\right)^2.$$

4 Why Automation is Necessary

Scientists cannot manually inspect millions of stars.

Modern telescopes (Kepler, TESS):

- Monitor thousands of stars simultaneously.
- Collect massive datasets.
- Require automated detection algorithms.

This is where simple machine learning logic helps.

5 Google Colab Notebook

We used the following live notebook:

https://colab.research.google.com/drive/17rF8bVm689Nq1yxqGVNw1xBzyJzTk0U1?usp=sharing

The notebook downloads real TESS light curve data and identifies potential transit dips.

6 Code Overview

6.1 Installing Required Library

```
!pip install lightkurve --quiet
```

- `lightkurve` is a Python package for analyzing Kepler and TESS data.

6.2 Importing Libraries

```
import lightkurve as lk
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import find_peaks
```

- `lightkurve` → download and handle light curves.
- `numpy` → numerical operations.
- `matplotlib` → plotting.
- `find`$_{p}eaks\text{ß}detectdipsautomatically$.

6.3 Downloading TESS Data

```
star_name = "HD 209458"
lc_all = lk.search_lightcurve(star_name, mission="TESS").download_all()
```

- HD 209458 is a well-known exoplanet host star.
- We download all available TESS observations.

6.4 Cleaning and Normalizing Data

```
lc = lc.normalize().remove_nans()
time = lc.time.value
flux = lc.flux.value
```

- Normalize brightness to 1.
- Remove missing values.
- Extract time and brightness arrays.

7 Smoothing the Data

```
window = max(5, int(len(flux) * 0.001))
flux_smooth = np.convolve(flux,
    np.ones(window)/window, mode="same")
```

Purpose:

- Reduce noise.
- Make transit dips clearer.
- Apply moving-average smoothing.

8 Detecting Transit Dips

We invert the flux so dips become peaks:

```
flux_inverted = 1 - flux_smooth
peaks, _ = find_peaks(flux_inverted,
                      height=0.005,
                      distance=20,
                      prominence=0.0015)
transit_times = time[peaks]
```

Interpretation:

- `height` → minimum dip depth.
- `distance` → minimum spacing between detections.
- `prominence` → ensures dip stands out from noise.

This automatically finds candidate transits.

9 Orbital Period Estimation

Once we detect multiple transit times:

- Compute time differences between dips.
- Average spacing gives orbital period.

$$P \approx \langle t_{n+1} - t_n \rangle$$

This matches Kepler's third law relationships.

10 Physics Behind the Code

This program combines:

- Observational astronomy (TESS data)
- Signal processing (smoothing + peak detection)
- Physics (orbital motion)
- Basic machine learning logic (pattern recognition)

Even simple algorithms can detect real planets.

11 Big Picture

This meeting connects:

- Exoplanet physics
- Real astronomical data
- Programming
- Automation in modern astrophysics

Modern astronomy is data-driven.
Learning how to analyze data is just as important as understanding the physics.